

Machine Learning for Robot Sensor Data

Scenario-Based Examination Questions

15-Week Course (Weeks 1-15)

15 Weeks • 75 Scenario Questions • 20 Marks per Question

Instructions to candidates

1. Each scenario question carries **20 marks**.
2. Read each scenario carefully and answer all of its sub-questions.
3. Marks for each sub-question are shown in brackets. No sub-question carries more than 4 marks.
4. Use examples from the scenario wherever possible.

Weeks

Week 1: From Robot Sensors to Data — The Four Types of Analysis
Week 2: Choosing the Learning Type and Preparing Sensor Data
Week 3: Linear Regression
Week 4: Logistic Regression and Its Cost Function
Week 5: Decision Trees
Week 6: k-Nearest Neighbours and Naive Bayes
Week 7: Support Vector Machines (SVM)
Week 8: Ensemble Learning and Random Forests
Week 9: Model Evaluation and Optimisation
Week 10: Unsupervised Learning I — Clustering
Week 11: Dimensionality Reduction, Anomaly Detection and Association Rules
Week 12: Reinforcement Learning
Week 13: Semi-Supervised and Self-Supervised Learning
Week 14: Time Series and Forecasting
Week 15: Deployment and MLOps — From Notebook to Robot Fleet

Week 1: From Robot Sensors to Data — The Four Types of Analysis

Question 1.1: The Warehouse Manager's Questions

A warehouse manager sends the data team four questions about the RoboFleet robots: (i) "How many pallets did each robot move in May?" (ii) "Why did AMR-04's battery life fall last month?" (iii) "Will AMR-11's motor fail this week?" (iv) "Which robot should take the next urgent order?"

- a)** Explain the four types of analysis and the question each answers. **(4 marks)**
- b)** Classify each of the manager's four questions, identifying the clue words you used. **(4 marks)**
- c)** Explain why the four types are described as a ladder. **(3 marks)**
- d)** Describe three tools used in diagnostic analysis. **(3 marks)**
- e)** Define a confounder. **(2 marks)**
- f)** Name two outputs of descriptive analysis. **(2 marks)**
- g)** Explain the mistake of answering a prescriptive question with a descriptive chart. **(2 marks)**

[Total: 20 marks]

Question 1.2: The Hot Motor Readings

AMR-09 logged five motor temperatures: 48, 50, 52, 54 and 86 °C.

- a)** Calculate the mean. **(3 marks)**
- b)** Find the median. **(3 marks)**
- c)** Calculate the standard deviation. Show your working. **(3 marks)**
- d)** Explain which of the mean and median better describes a typical reading, and why. **(3 marks)**
- e)** Calculate the range and comment on what it shows. **(3 marks)**
- f)** Write the diagnostic question the 86 °C reading should prompt. **(3 marks)**
- g)** Define a percentile. **(2 marks)**

[Total: 20 marks]

Question 1.3: From Thermistor to Table

A new robot model is joining the fleet. Its motor thermistor, accelerometer, load cell, camera and BMS all produce data that must end up in the fleet's analysis table.

- a)** Describe the five steps that turn a thermistor reading into a stored, summarised row. **(4 marks)**

- b)** Classify four of the new robot's data streams by data kind (continuous, discrete, categorical, binary, time series, image). **(4 marks)**
- c)** Explain why the edge computer turns 60 one-second readings into one-minute features. **(4 marks)**
- d)** Explain what each row and each column of the analysis table represents. **(4 marks)**
- e)** Explain how descriptive analysis reveals data problems before any model is built. **(4 marks)**

[Total: 20 marks]

Question 1.4: Faults on the Ramps

The fleet fault rate is 5%. Broken down by floor type it is tile 2%, concrete 4% and ramp 12%. Motor temperature and faults have a correlation of $r = 0.6$, but faults also rise on hot afternoons.

- a)** Define drill-down. **(2 marks)**
- b)** Interpret the floor-type breakdown. **(3 marks)**
- c)** Interpret $r = 0.6$ and explain the full range of correlation values. **(4 marks)**
- d)** Explain why correlation is not causation. **(3 marks)**
- e)** Explain how ambient temperature could act as a confounder here. **(4 marks)**
- f)** Propose the next steps in this diagnostic analysis. **(4 marks)**

[Total: 20 marks]

Question 1.5: The Eager Data Scientist

A new data scientist wants to train a deep learning model on RoboFleet data on her first day, without looking at the data first.

- a)** Explain why descriptive analysis should come before machine learning. **(4 marks)**
- b)** Explain how diagnostic analysis guides feature selection. **(3 marks)**
- c)** Identify which analysis types machine learning mainly powers. **(3 marks)**
- d)** Write one question of each analysis type about the fleet's batteries. **(4 marks)**
- e)** Define an observation in a dataset. **(2 marks)**
- f)** Propose a plan for her first week of analysis. **(4 marks)**

[Total: 20 marks]

Week 2: Choosing the Learning Type and Preparing Sensor Data

Question 2.1: Five Problems, Five Learning Types

The team has five projects: predicting faults from labelled records; finding operating modes in unlabelled logs; learning when to charge by trial and error; classifying grasps with 150 labels and 1 million unlabelled recordings; and pre-training a model by predicting the next second of IMU data.

- a) Explain the five learning types. (4 marks)
- b) Describe the order of questions in the decision guide for choosing a learning type. (4 marks)
- c) Justify semi-supervised learning for the grasp project. (3 marks)
- d) Justify reinforcement learning for the charging project. (3 marks)
- e) Explain the quick test for telling classification from regression. (2 marks)
- f) Classify "predict battery SoC in 10 minutes" as classification or regression. (2 marks)
- g) Classify "predict the floor type" as classification or regression. (2 marks)

[Total: 20 marks]

Question 2.2: Cleaning the Logs

A day of RoboFleet data contains motor temperatures of -40°C in a warm warehouse, gaps caused by Wi-Fi dropouts, and sudden vibration spikes. The `floor_type` column holds text values.

- a) Explain how to handle the -40°C readings. (3 marks)
- b) Explain three options for handling the missing values. (3 marks)
- c) Explain how to decide whether to keep or remove the vibration spikes. (3 marks)
- d) Explain how one-hot encoding converts `floor_type`. (3 marks)
- e) Explain why floor types must not be coded as 1, 2 and 3. (3 marks)
- f) Calculate the RMS and the plain mean of the readings 0.5, -0.5 , 0.5, -0.5 g, and explain the difference. (3 marks)
- g) Define feature engineering. (2 marks)

[Total: 20 marks]

Question 2.3: Putting Features on the Same Scale

Payload has a mean of 20 kg and a standard deviation of 8 kg. Motor current ranges from 2 A to 8 A. The team plans to use k-NN.

- a)** Explain why distance-based methods need scaled features. **(4 marks)**
- b)** Standardise a payload of 36 kg. **(4 marks)**
- c)** Min-max normalise a motor current of 5 A. **(4 marks)**
- d)** Compare standardisation and min-max normalisation. **(4 marks)**
- e)** Explain why the scaler must be fitted on training data only. **(4 marks)**

[Total: 20 marks]

Question 2.4: The 99% Model

A student shuffled 12 months of data from 20 robots, split it randomly, and reported 99% accuracy for fault prediction. The supervisor is suspicious.

- a)** Define data leakage. **(2 marks)**
- b)** Explain the roles and typical sizes of the training, validation and test sets. **(3 marks)**
- c)** Explain time leakage. **(4 marks)**
- d)** Explain robot leakage. **(3 marks)**
- e)** Explain why the supervisor is right to be suspicious. **(4 marks)**
- f)** Propose a fair way to split the data. **(4 marks)**

[Total: 20 marks]

Question 2.5: Linking Analysis and Learning

A lecturer asks students to connect the four analysis types from Week 1 to the learning types from Week 2.

- a)** Explain how descriptive analysis links to unsupervised learning. **(4 marks)**
- b)** Explain which learning types support diagnostic analysis. **(3 marks)**
- c)** Explain which learning types support predictive analysis. **(3 marks)**
- d)** Explain how prescriptive analysis is achieved. **(4 marks)**
- e)** Explain the "cover half a sentence" analogy for self-supervised learning. **(2 marks)**
- f)** For three robot questions of your choice, state the analysis type and learning type. **(4 marks)**

[Total: 20 marks]

Week 3: Linear Regression

Question 3.1: Speed and Motor Current

The team records robot speed x (m/s) and motor current y (A): (0.5, 2.1), (1.0, 2.4), (1.5, 3.0), (2.0, 3.5), (2.5, 4.0).

- a) Explain each term in $\hat{y} = \beta_0 + \beta_1 x$. (4 marks)
- b) Calculate the slope β_1 using least squares. Show your working. (4 marks)
- c) Calculate the intercept β_0 . (3 marks)
- d) Predict the current at 1.8 m/s. (3 marks)
- e) Interpret the slope. (2 marks)
- f) Interpret the intercept. (2 marks)
- g) Explain the caution needed when predicting the current at 4 m/s. (2 marks)

[Total: 20 marks]

Question 3.2: Checking the Fit

A fitted model is $\text{current} = 1.5 + 0.1 \times \text{payload}$. Three test observations are: 20 kg $\rightarrow$ 3.8 A, 30 kg $\rightarrow$ 4.4 A, 40 kg $\rightarrow$ 5.9 A. Separately, a robot carrying 25 kg draws 6.1 A.

- a) Calculate the residual for each test observation. (3 marks)
- b) Calculate the mean squared error. (3 marks)
- c) Explain why errors are squared in MSE. (3 marks)
- d) Use the model to diagnose the robot drawing 6.1 A at 25 kg. (3 marks)
- e) Explain gradient descent using the foggy-hill analogy. (3 marks)
- f) Explain what happens if the learning rate is too large or too small. (3 marks)
- g) Define R^2 . (2 marks)

[Total: 20 marks]

Question 3.3: Multiple Linear Regression

A fitted RoboFleet model is: $\text{current} = 1.0 + 0.09 \times \text{payload} + 1.2 \times \text{speed} + 0.7 \times \text{ramp} - 0.005 \times \text{battery_soc}$.

- a) Predict the current for 40 kg, 1.2 m/s, on a ramp, at 80% SoC. (4 marks)
- b) Predict the current for 10 kg, 0.8 m/s, on tile, at 40% SoC. (4 marks)
- c) Interpret the payload and speed coefficients. (4 marks)
- d) Explain why features must be standardised before comparing coefficients. (4 marks)

- e) Interpret the ramp coefficient and explain how it relates to one-hot encoding. (4 marks)

[Total: 20 marks]

Question 3.4: When a Straight Line Fails

The team finds that motor temperature rises with the square of current, that motor current and motor power are both in the feature set, and that residuals grow larger at high payloads.

- a) State the linearity assumption. (2 marks)
- b) Explain how to handle the square relationship. (3 marks)
- c) Explain multicollinearity using current and power. (4 marks)
- d) Explain the constant-spread assumption and the growing residuals. (3 marks)
- e) Explain why independent errors are a concern with time-series sensor data. (4 marks)
- f) Explain how residual plots help detect these problems. (4 marks)

[Total: 20 marks]

Question 3.5: Reading the Model Carefully

A manager says the ramp coefficient "proves ramps damage motors" and wants to predict the current for a 150 kg load, although training data only went up to 50 kg. The team also wants to flag any minute where current exceeds the prediction by more than 1.5 A.

- a) Explain why a coefficient shows association rather than cause. (4 marks)
- b) Explain the risk of the 150 kg prediction. (3 marks)
- c) Explain how uncleaned outliers could distort the model. (3 marks)
- d) Explain how the 1.5 A residual flags support maintenance. (4 marks)
- e) Interpret an R^2 of 0.82. (2 marks)
- f) Propose how the team should use the model responsibly. (4 marks)

[Total: 20 marks]

Week 4: Logistic Regression and Its Cost Function

Question 4.1: Will the Grasp Hold?

A grasp model is $z = -8 + 1.0 \times \text{grip_force (N)}$, with $p = 1 \div (1 + e^{-z})$.

- a) Explain why linear regression is unsuitable for a 0/1 label. (4 marks)
- b) Calculate the probability of success at 10 N. (4 marks)
- c) Calculate the probability of success at 6 N. (3 marks)
- d) Find the force at the decision boundary. (3 marks)
- e) Define the decision threshold. (2 marks)
- f) Calculate how much each extra newton multiplies the odds of success. (2 marks)
- g) Find the force needed for a 95% success probability ($\sigma(2.94) \approx 0.95$). (2 marks)

[Total: 20 marks]

Question 4.2: Calculating Log Loss

A model makes three predictions: truth 1 with $p = 0.9$; truth 0 with $p = 0.2$; truth 1 with $p = 0.3$.

- a) Calculate the log loss for the first prediction. (3 marks)
- b) Calculate the log loss for the second prediction. (3 marks)
- c) Calculate the log loss for the third prediction. (3 marks)
- d) Calculate the average log loss. (3 marks)
- e) Explain why logistic regression uses log loss instead of MSE. (3 marks)
- f) Explain what a convex cost surface means for gradient descent. (3 marks)
- g) Identify which prediction is penalised most and explain why. (2 marks)

[Total: 20 marks]

Question 4.3: The Rare-Fault Model

Only 3% of robot-days end in a fault. A logistic regression fault model uses four standardised features.

- a) Calculate $\sigma(0)$, $\sigma(3)$ and $\sigma(-1)$. (4 marks)
- b) Explain the effect of lowering the threshold from 0.5 to 0.3. (4 marks)
- c) Explain class imbalance and how `class_weight="balanced"` helps. (4 marks)
- d) Calculate the accuracy of a model that always predicts "no fault" and explain why it is misleading. (4 marks)
- e) Explain regularisation and the role of C. (4 marks)

[Total: 20 marks]

Question 4.4: Tile, Concrete or Ramp?

A logistic model must classify floor type from vibration and current features. For one reading it outputs tile 0.6, concrete 0.3, ramp 0.1.

- a) Define multi-class classification. (2 marks)
- b) Explain the one-vs-rest approach. (3 marks)
- c) Explain softmax (multinomial) regression. (4 marks)
- d) Explain why features should be scaled. (3 marks)
- e) Interpret the output probabilities and state the prediction. (4 marks)
- f) Design an experiment to build and test the floor-type classifier. (4 marks)

[Total: 20 marks]

Question 4.5: Gripping a Fragile Object

For a fragile object, the grasp model is $z = -5 + 0.6 \times \text{grip_force}$. The object is damaged above 12 N. Use $\sigma(2.94) \approx 0.95$ and $\sigma(2.20) \approx 0.90$.

- a) Calculate the force needed for 95% success. (4 marks)
- b) Compare this force with the damage limit. (3 marks)
- c) Calculate the force needed for 90% success. (3 marks)
- d) Recommend a grip force and explain the trade-offs. (4 marks)
- e) Calculate the odds multiplier per extra newton. (2 marks)
- f) Suggest other ways to improve grasp success without exceeding 12 N. (4 marks)

[Total: 20 marks]

Week 5: Decision Trees

Question 5.1: Measuring Impurity

A tree node contains 12 robot-days: 9 with a fault and 3 without.

- a) Describe the parts of a decision tree. (4 marks)
- b) Calculate the Gini impurity of the node. (4 marks)
- c) Calculate the entropy of the node. (3 marks)
- d) State the Gini and entropy values of a pure node and explain why. (3 marks)
- e) State the maximum Gini and entropy for two classes. (2 marks)
- f) Define information gain. (2 marks)
- g) Define tree depth. (2 marks)

[Total: 20 marks]

Question 5.2: Choosing the Best Split

A node has 10 robot-days: 5 faults and 5 OK. Split A ($\text{vibration_rms} > 0.45 \text{ g}$) gives Yes: 4 faults, 0 OK; No: 1 fault, 5 OK. Split B ($\text{motor_temp} > 70 \text{ }^\circ\text{C}$) gives Yes: 3 faults, 2 OK; No: 2 faults, 3 OK.

- a) Calculate the Gini impurity before splitting. (3 marks)
- b) Calculate the Gini impurity of each branch of split A. (3 marks)
- c) Calculate the weighted Gini and the Gini decrease for split A. (3 marks)
- d) Calculate the weighted Gini and the Gini decrease for split B. (3 marks)
- e) Choose the better split and justify it. (3 marks)
- f) Explain what the tree does next. (3 marks)
- g) State the entropy of the node before splitting. (2 marks)

[Total: 20 marks]

Question 5.3: Four Tree Algorithms

A student is comparing ID3, C4.5, CART and CHAID for a maintenance project with numeric sensor features and a categorical fault label.

- a) Explain ID3 and its bias. (4 marks)
- b) Explain how C4.5 improves on ID3. (4 marks)
- c) Explain CART. (4 marks)
- d) Explain CHAID. (4 marks)
- e) Identify which algorithm scikit-learn uses and explain its implications. (4 marks)

[Total: 20 marks]

Question 5.4: The Perfect Training Score

An unlimited decision tree scores 100% on training data but only 71% on test data.

- a)** Define overfitting. **(2 marks)**
- b)** Explain why an unlimited tree overfits. **(3 marks)**
- c)** Explain how `max_depth` and `min_samples_leaf` reduce overfitting. **(4 marks)**
- d)** Explain cost-complexity pruning. **(3 marks)**
- e)** Design an experiment to find a suitable tree depth. **(4 marks)**
- f)** Explain two strengths and two weaknesses of trees to a maintenance technician. **(4 marks)**

[Total: 20 marks]

Question 5.5: Predicting Remaining Useful Life

A regression tree predicts remaining useful life (RUL). One leaf contains four robots with RUL of 500, 540, 620 and 580 hours.

- a)** Explain how a regression tree chooses its splits. **(4 marks)**
- b)** Calculate the leaf's prediction. **(3 marks)**
- c)** Explain why a regression tree's predictions form a step function. **(3 marks)**
- d)** Explain why a regression tree cannot predict beyond the range of its training labels. **(4 marks)**
- e)** Explain why feature importance is not proof of cause. **(2 marks)**
- f)** Compare a regression tree with linear regression for predicting RUL. **(4 marks)**

[Total: 20 marks]

Week 6: k-Nearest Neighbours and Naive Bayes

Question 6.1: Classifying the Floor with k-NN

A new standardised reading is at (1, 1). Stored examples are: A (0, 1) tile; B (1, 0) tile; C (2, 1) ramp; D (2, 2) ramp; E (1, 3) concrete.

- a) Explain how k-NN works and why it is called a lazy learner. (4 marks)
- b) Calculate the Euclidean distance from the new reading to each example. (4 marks)
- c) Predict the floor type with $k = 3$. (3 marks)
- d) Predict with $k = 5$, first by simple vote and then by $1 \div \text{distance}$ weighting. (3 marks)
- e) Define a decision boundary. (2 marks)
- f) Explain why an odd k is preferred for two classes. (2 marks)
- g) Explain the curse of dimensionality. (2 marks)

[Total: 20 marks]

Question 6.2: Tuning k-NN

A k-NN floor classifier uses payload (0–50 kg) and vibration (0.1–0.6 g) without scaling. Accuracy is poor, and predictions take a long time on the full dataset.

- a) Explain why scaling is essential for k-NN. (3 marks)
- b) Explain how payload would dominate vibration without scaling. (3 marks)
- c) Explain the effect of a very small k . (3 marks)
- d) Explain the effect of a very large k . (3 marks)
- e) Explain how to choose k . (3 marks)
- f) Explain why k-NN prediction is slow on large datasets. (3 marks)
- g) Explain how k-NN performs regression. (2 marks)

[Total: 20 marks]

Question 6.3: Updating the Fault Risk

5% of robot-days end in a fault. "High temperature" appears on 90% of fault days and 15% of normal days. "High vibration" appears on 80% of fault days and 10% of normal days.

- a) Explain the prior, likelihood and posterior in Bayes' theorem. (4 marks)
- b) Calculate $P(\text{hot})$. (4 marks)
- c) Calculate $P(\text{fault} \mid \text{hot})$. (4 marks)

- d)** Calculate $P(\text{fault} \mid \text{hot, high vibration})$ using the naive assumption. **(4 marks)**
- e)** Explain the naive assumption and whether it holds here. **(4 marks)**

[Total: 20 marks]

Question 6.4: Three Naive Bayes Variants

The team has three datasets: continuous sensor readings; hourly counts of log codes such as OVERCURRENT and SLIP; and records of which warning codes appeared at all.

- a)** Define likelihood. **(2 marks)**
- b)** Explain Gaussian Naive Bayes. **(3 marks)**
- c)** Explain Multinomial Naive Bayes. **(4 marks)**
- d)** Explain Bernoulli Naive Bayes. **(3 marks)**
- e)** Explain Laplace smoothing and why it is needed. **(4 marks)**
- f)** Match each dataset to a Naive Bayes variant and justify each. **(4 marks)**

[Total: 20 marks]

Question 6.5: A Model for the Robot's Microcontroller

A fault classifier must run on a robot's small edge computer with limited memory, using a dataset of 2 million rows and 15 features.

- a)** Compare k-NN and Naive Bayes in training and prediction speed. **(4 marks)**
- b)** Compare their need for scaling. **(3 marks)**
- c)** Compare how they handle many features. **(3 marks)**
- d)** Compare the probabilities they produce. **(4 marks)**
- e)** Define a lazy learner. **(2 marks)**
- f)** Recommend a method for the edge computer and justify it. **(4 marks)**

[Total: 20 marks]

Week 7: Support Vector Machines (SVM)

Question 7.1: Classifying Bearings

A trained linear SVM on standardised features has $w = (1, 2)$ and $b = -4$. Class +1 is damaged; -1 is healthy. Three bearings are tested: A (2, 1), B (3, 2) and C (1, 1).

- a) Explain a hyperplane and the maximum margin. (4 marks)
- b) Calculate $w \cdot x + b$ for each bearing and classify it. (4 marks)
- c) Identify which bearing lies on a margin edge and explain what this means. (3 marks)
- d) Calculate the margin width. (3 marks)
- e) Define a support vector. (2 marks)
- f) Explain what the size of $w \cdot x + b$ indicates. (2 marks)
- g) Explain why features must be scaled for an SVM. (2 marks)

[Total: 20 marks]

Question 7.2: Calculating Hinge Loss

A soft-margin SVM scores three training points: P1 ($y = +1$, score 0.4), P2 ($y = +1$, score 1.5) and P3 ($y = -1$, score 0.2).

- a) Calculate the hinge loss for P1. (3 marks)
- b) Calculate the hinge loss for P2. (3 marks)
- c) Calculate the hinge loss for P3. (3 marks)
- d) Interpret the result for P3. (3 marks)
- e) Explain the soft margin. (3 marks)
- f) Explain the effect of small and large C values. (3 marks)
- g) State the hinge loss formula. (2 marks)

[Total: 20 marks]

Question 7.3: Curved Boundaries

A motor's vibration frequency offset is near 0 when healthy, but faults appear both far below and far above 0. No straight cut separates healthy from faulty.

- a) Explain why no single threshold separates the classes. (4 marks)
- b) Explain how adding x^2 as a feature makes the data separable. (4 marks)
- c) Explain the kernel trick. (4 marks)
- d) Calculate the RBF similarity of two points with squared distance 4, for $\gamma = 0.25$ and for $\gamma = 1$. (4 marks)

- e) Explain how γ affects the shape of the boundary. (4 marks)

[Total: 20 marks]

Question 7.4: Two Million Rows

A student tries to train an RBF SVM on 2 million raw one-second rows. Training has run for hours, and the maintenance team also wants fault probabilities.

- a) State one weakness of SVMs revealed here. (2 marks)
- b) Explain why SVM training is slow on this dataset. (3 marks)
- c) Explain how summarising the data into windows helps. (4 marks)
- d) Explain when a linear SVM is a better alternative. (3 marks)
- e) Explain why SVMs do not give probabilities directly and how this is addressed. (4 marks)
- f) Propose a plan to build the bearing classifier efficiently. (4 marks)

[Total: 20 marks]

Question 7.5: Choosing and Tuning the Kernel

A team is building an SVM bearing classifier with three spectral features and about 5,000 labelled windows. It must also predict a continuous wear level.

- a) Explain when to use a linear kernel and when to use an RBF kernel. (4 marks)
- b) Explain the polynomial kernel. (3 marks)
- c) Explain how C and γ should be tuned. (3 marks)
- d) Discuss the interpretability of SVMs with non-linear kernels. (4 marks)
- e) Explain the ϵ -tube in support vector regression. (2 marks)
- f) Recommend an SVM setup for the team and justify it. (4 marks)

[Total: 20 marks]

Week 8: Ensemble Learning and Random Forests

Question 8.1: The Wisdom of Crowds

Three independent fault classifiers are each correct 80% of the time. They are combined by majority vote.

- a) Explain why combining models can improve accuracy. (4 marks)
- b) Calculate the probability that the majority vote is correct. (4 marks)
- c) Explain why independence between models is essential. (3 marks)
- d) Explain three ways to create diversity in an ensemble. (3 marks)
- e) Define an ensemble. (2 marks)
- f) Distinguish between voting and stacking. (2 marks)
- g) Define a meta-model. (2 marks)

[Total: 20 marks]

Question 8.2: Building a Random Forest

A random forest uses 16 features. Four regression trees predict RUL of 420, 380, 400 and 440 hours. In a 300-tree classifier, 210 trees vote "fault".

- a) Explain how a bootstrap sample is drawn. (3 marks)
- b) Explain out-of-bag rows and their use. (3 marks)
- c) Explain the random feature subset used at each split. (3 marks)
- d) Calculate how many features each split considers using the $\sqrt{p}$ rule. (3 marks)
- e) Calculate the forest's RUL prediction from the four trees. (3 marks)
- f) Calculate the classifier's fault probability. (3 marks)
- g) State a typical starting value for $n_{\text{estimators}}$. (2 marks)

[Total: 20 marks]

Question 8.3: One Round of AdaBoost

An AdaBoost stump misclassifies 25% of the weighted training rows.

- a) Describe the steps of AdaBoost. (4 marks)
- b) Calculate the stump's vote strength α . (4 marks)
- c) Calculate the weight multipliers for misclassified and correct rows. (4 marks)
- d) Explain why AdaBoost focuses on hard cases and why this helps with fault data. (4 marks)
- e) Calculate α when $\varepsilon = 0.5$ and explain what it means. (4 marks)

[Total: 20 marks]

Question 8.4: Gradient Boosting for RUL

A gradient-boosting model starts by predicting 350 hours for every robot. AMR-06 truly has 450 hours left. Tree 1 predicts a residual of +90 hours for AMR-06, and the learning rate is 0.1.

- a)** Define a residual. **(2 marks)**
- b)** Calculate the new prediction after tree 1. **(3 marks)**
- c)** Explain why a small learning rate with many trees generalises better. **(4 marks)**
- d)** Calculate the new residual and explain what tree 2 learns. **(3 marks)**
- e)** Explain four features that make XGBoost effective. **(4 marks)**
- f)** Explain why boosting can overfit and how early stopping helps. **(4 marks)**

[Total: 20 marks]

Question 8.5: Bagging or Boosting?

A maintenance team must choose between a random forest and XGBoost. Some fault labels are known to be noisy, the team has little time for tuning, and technicians want to understand the model.

- a)** Compare bagging and boosting in how they train and what they fix. **(4 marks)**
- b)** Compare their sensitivity to noisy labels. **(3 marks)**
- c)** Compare the tuning effort each needs. **(3 marks)**
- d)** Compare impurity-based and permutation feature importance. **(4 marks)**
- e)** Name two gradient-boosting libraries other than XGBoost. **(2 marks)**
- f)** Recommend an approach and explain how to keep technicians' trust. **(4 marks)**

[Total: 20 marks]

Week 9: Model Evaluation and Optimisation

Question 9.1: The Confusion Matrix

A fault model is tested on 2,000 robot-days containing 150 real faults. It gives $TP = 120$, $FN = 30$, $FP = 100$ and $TN = 1,750$.

- a) Explain the four cells of the confusion matrix in robot terms. (4 marks)
- b) Calculate accuracy and precision. (4 marks)
- c) Calculate recall and specificity. (3 marks)
- d) Calculate the F1-score. (3 marks)
- e) Calculate the accuracy of a model that always predicts "no fault". (2 marks)
- f) Define a false negative. (2 marks)
- g) Define a false positive. (2 marks)

[Total: 20 marks]

Question 9.2: Choosing a Threshold from the ROC Table

A fault model gives these results: threshold 0.8 $\rightarrow$ TPR 0.55, FPR 0.01; 0.5 $\rightarrow$ TPR 0.78, FPR 0.04; 0.3 $\rightarrow$ TPR 0.90, FPR 0.10; 0.1 $\rightarrow$ TPR 0.97, FPR 0.30.

- a) Define TPR and FPR. (3 marks)
- b) Explain what happens as the threshold is lowered. (3 marks)
- c) Choose a threshold if the false-positive rate must stay at or below 5%. (3 marks)
- d) Choose a threshold if recall must be at least 0.9. (3 marks)
- e) Explain what AUC measures. (3 marks)
- f) Explain what an AUC of 0.5 means and when a precision-recall curve is better. (3 marks)
- g) State the axes of an ROC curve. (2 marks)

[Total: 20 marks]

Question 9.3: Measuring RUL Predictions

For three motors, the actual RUL is 400, 300 and 200 hours, and the predictions are 380, 330 and 190 hours.

- a) Calculate the MAE. (4 marks)
- b) Calculate the RMSE. (4 marks)
- c) Calculate the MAPE. (4 marks)
- d) Calculate R^2 . (4 marks)

- e) Recommend which metric to report to maintenance planners and explain why MAPE fails when true values are near 0. (4 marks)

[Total: 20 marks]

Question 9.4: Choosing a Cross-Validation Scheme

The team faces three situations: faults occur on only 4% of rows; the model must work on robots it has never seen; and the data is a time series of sensor readings.

- a) Define cross-validation. (2 marks)
- b) Explain stratified k-fold. (3 marks)
- c) Explain group k-fold. (4 marks)
- d) Explain time-series split. (3 marks)
- e) Explain why the final test set must stay untouched. (4 marks)
- f) Match each situation to a cross-validation scheme and justify each. (4 marks)

[Total: 20 marks]

Question 9.5: Depth, Bias and Variance

A tree's training/validation accuracy is 0.70/0.69 at depth 2, 0.88/0.86 at depth 6, and 0.99/0.74 at depth 20.

- a) Interpret the results at each depth. (4 marks)
- b) Explain bias and variance. (3 marks)
- c) Compare Ridge and Lasso regularisation. (3 marks)
- d) Compare grid search, random search and Bayesian optimisation. (4 marks)
- e) Distinguish between a parameter and a hyperparameter. (2 marks)
- f) Explain the criteria for selecting the final model. (4 marks)

[Total: 20 marks]

Week 10: Unsupervised Learning I — Clustering

Question 10.1: K-Means by Hand

Motor currents (A) are 1.8, 2.0, 2.2, 5.8, 6.0 and 7.0. Use $k = 2$ with starting centroids $c_1 = 1.8$ and $c_2 = 2.4$.

- a) Describe the steps of the K-Means algorithm. (4 marks)
- b) Carry out the first assignment and centroid update. (4 marks)
- c) Carry out the second iteration and state the final centroids. (3 marks)
- d) Interpret the two clusters as operating modes. (3 marks)
- e) Define WCSS. (2 marks)
- f) Explain why features must be scaled before clustering. (2 marks)
- g) Explain how K-Means++ chooses starting centroids. (2 marks)

[Total: 20 marks]

Question 10.2: How Many Operating Modes?

K-Means on RoboFleet data gives WCSS values of 1,200 ($k = 1$), 600 ($k = 2$), 520 ($k = 3$), 480 ($k = 4$) and 460 ($k = 5$).

- a) Explain the elbow method. (3 marks)
- b) Choose k from these results and justify it. (3 marks)
- c) Explain the silhouette score. (3 marks)
- d) Explain why domain knowledge should also guide the choice of k . (3 marks)
- e) Explain why WCSS always falls as k increases. (3 marks)
- f) Explain why each cluster must be interpreted. (3 marks)
- g) State the range of the silhouette score. (2 marks)

[Total: 20 marks]

Question 10.3: Grouping the Twenty Robots

The fleet manager wants to group the 20 robots by their weekly usage profiles to plan maintenance batches, without deciding the number of groups in advance.

- a) Explain how agglomerative clustering works. (4 marks)
- b) Explain how to read and cut a dendrogram. (4 marks)
- c) Compare single, complete and average linkage. (4 marks)
- d) Explain Ward linkage. (4 marks)
- e) Explain why hierarchical clustering is unsuitable for very large datasets. (4 marks)

[Total: 20 marks]

Question 10.4: Obstacles in the LiDAR Scan

A robot's LiDAR returns points from a pallet, a person and a wall, plus stray noise. Some obstacles are much denser than others.

- a)** Explain the `eps` and `min_samples` settings. **(2 marks)**
- b)** Explain core, border and noise points. **(3 marks)**
- c)** Explain why DBSCAN suits LiDAR obstacle clustering. **(4 marks)**
- d)** Explain why noise points are valuable in sensor data. **(3 marks)**
- e)** Explain DBSCAN's weakness with varying densities and how OPTICS helps. **(4 marks)**
- f)** Propose a plan to tune DBSCAN for the robot. **(4 marks)**

[Total: 20 marks]

Question 10.5: The In-Between Reading

A motor current of 4.0 A lies between a light-load mode (mean 2.1 A) and a heavy-load mode (mean 6.3 A). The team is considering a Gaussian mixture model.

- a)** Compare a GMM with K-Means. **(4 marks)**
- b)** Explain soft membership using the 4.0 A reading. **(3 marks)**
- c)** Explain the E-step of the EM algorithm. **(3 marks)**
- d)** Explain the M-step of the EM algorithm. **(4 marks)**
- e)** Explain how BIC helps choose the number of components. **(2 marks)**
- f)** Recommend a clustering method for each of: round clusters in large data, odd shapes with noise, and overlapping groups. **(4 marks)**

[Total: 20 marks]

Week 11: Dimensionality Reduction, Anomaly Detection and Association Rules

Question 11.1: Compressing Vibration Features

PCA on five standardised motor features gives eigenvalues of 5.0, 2.5, 1.5, 0.6 and 0.4.

- a) Explain the curse of dimensionality. (4 marks)
- b) Describe the steps of PCA. (4 marks)
- c) Calculate the explained variance of PC1, PC2 and PC3. (3 marks)
- d) Determine how many components are needed to keep at least 90% of the variance. (3 marks)
- e) Suggest a name for PC1 if it loads heavily on current, temperature and vibration. (2 marks)
- f) Explain why features must be standardised before PCA. (2 marks)
- g) State how PC2 relates to PC1. (2 marks)

[Total: 20 marks]

Question 11.2: Beyond PCA

A team has four tasks: visualising operating modes in 2D; splitting vibration spectra into motor hum, gear mesh and bearing tone; separating two vibration sources recorded by one accelerometer; and analysing robot-arm joint trajectories that lie on a curved surface.

- a) Explain t-SNE. (3 marks)
- b) Explain why t-SNE outputs should not be used as model inputs. (3 marks)
- c) Explain NMF and why it suits vibration spectra. (3 marks)
- d) Explain ICA and the "cocktail party" problem. (3 marks)
- e) Explain Isomap. (3 marks)
- f) Match each task to a method. (3 marks)
- g) Identify which of these methods are linear. (2 marks)

[Total: 20 marks]

Question 11.3: Is This Reading Unusual?

Normal vibration_rms has $\mu = 0.25$ g and $\sigma = 0.04$ g. Two new readings are 0.39 g and 0.33 g. The first and third quartiles are 0.22 g and 0.30 g.

- a) Calculate the z-score of each reading and decide whether each is an anomaly. (4 marks)
- b) Calculate the IQR upper and lower bounds. (4 marks)

- c) Explain how Isolation Forest detects anomalies. (4 marks)
- d) Explain how reconstruction error detects anomalies. (4 marks)
- e) Explain mode-aware anomaly detection and why it matters. (4 marks)

[Total: 20 marks]

Question 11.4: Events Before a Stall

In 2,000 shifts, {wheel_slip, ramp_use} occurs in 200, motor_stall occurs in 160, and all three occur together in 80.

- a) Define support. (2 marks)
- b) Calculate the support of the full rule. (3 marks)
- c) Calculate the confidence of {wheel_slip, ramp_use} $\Rightarrow$ {motor_stall}. (4 marks)
- d) Calculate the lift. (3 marks)
- e) Interpret the rule and explain what it does and does not prove. (4 marks)
- f) Explain why a rule with high confidence but lift near 1 is not useful. (4 marks)

[Total: 20 marks]

Question 11.5: Mining Five Years of Event Logs

The team wants to mine five years of shift event logs for frequent itemsets and rules.

- a) Explain the Apriori principle. (4 marks)
- b) Explain how Apriori builds itemsets level by level. (3 marks)
- c) Explain the FP-tree. (3 marks)
- d) Explain why FP-Growth is faster than Apriori. (4 marks)
- e) Explain the effect of the min_support setting. (2 marks)
- f) Recommend an algorithm for the five-year logs and justify it. (4 marks)

[Total: 20 marks]

Week 12: Reinforcement Learning

Question 12.1: The Charging Decision

A robot must decide when to go to the charger. It earns +1 per completed task and -20 if its battery runs flat.

- a) Describe the five parts of an MDP for this problem. (4 marks)
- b) Explain the Markov property and how state design supports it. (4 marks)
- c) Explain the return and the role of γ . (3 marks)
- d) Calculate the return for rewards +1, +1, -20 with $\gamma = 0.9$. (3 marks)
- e) Define a policy. (2 marks)
- f) Explain how a badly designed reward could be "gamed". (2 marks)
- g) Name two sensor readings that could form the state. (2 marks)

[Total: 20 marks]

Question 12.2: Value Iteration by Hand

In state "Queue", the action "dock" reaches the terminal state "Charged" with probability 0.9 (reward +5) and fails with probability 0.1, staying in "Queue" (reward -2). $\gamma = 0.9$ and $V(\text{Charged}) = 0$.

- a) Explain the Bellman equation in words. (3 marks)
- b) Calculate $Q(\text{Queue}, \text{dock})$ when $V(\text{Queue}) = 0$. (3 marks)
- c) Recalculate $Q(\text{Queue}, \text{dock})$ in the next sweep using your updated $V(\text{Queue})$. (3 marks)
- d) Explain when value iteration can be used. (3 marks)
- e) Explain how the optimal policy is derived. (3 marks)
- f) Explain the Q^* form of the Bellman equation. (3 marks)
- g) Explain why a terminal state has value 0. (2 marks)

[Total: 20 marks]

Question 12.3: Q-Learning versus SARSA

$Q(s, a) = 3$, $\alpha = 0.2$, $\gamma = 0.9$ and $r = 2$. In the next state, the best action has $Q = 6$, but the robot actually takes an exploratory action with $Q = 2$.

- a) Calculate the Q-learning update. (4 marks)
- b) Calculate the SARSA update. (4 marks)
- c) Explain on-policy and off-policy learning. (4 marks)
- d) Explain why SARSA tends to learn a safer path. (4 marks)

- e) Explain when a DQN is needed instead of a table. (4 marks)

[Total: 20 marks]

Question 12.4: Learning a Continuous Policy

A robot arm must learn continuous motor torques. The critic estimates $V(s) = 8$ and $V(s') = 9$; the action gave $r = 1$, with $\gamma = 0.9$.

- a) Define a policy-gradient method. (2 marks)
- b) Explain REINFORCE and its weakness. (3 marks)
- c) Calculate the advantage and explain what it means for the actor. (4 marks)
- d) Explain the roles of the actor and the critic. (3 marks)
- e) Explain how A3C works. (4 marks)
- f) Recommend an approach for learning continuous torques and justify it. (4 marks)

[Total: 20 marks]

Question 12.5: Deploying a Charging Policy Safely

A warehouse wants to deploy a learned charging policy on its fleet.

- a) Explain how sensor data forms the state and the reward. (4 marks)
- b) Explain why training usually starts in simulation. (3 marks)
- c) Explain offline RL. (3 marks)
- d) Explain why worst-case behaviour must be checked, not just average reward. (4 marks)
- e) Predict the effect of raising the flat-battery penalty from -20 to -100 . (2 marks)
- f) Propose a safe deployment plan. (4 marks)

[Total: 20 marks]

Week 13: Semi-Supervised and Self-Supervised Learning

Question 13.1: Few Labels, Many Recordings

After a month, the team has 300 labelled grasps and 60,000 unlabelled grasp recordings.

- a) Explain four reasons labels are scarce in robotics. (4 marks)
- b) Distinguish between semi-supervised and supervised learning. (4 marks)
- c) Explain the smoothness assumption with a grasp example. (3 marks)
- d) Explain the cluster assumption with a grasp example. (3 marks)
- e) Explain the manifold assumption. (2 marks)
- f) State when these assumptions break. (2 marks)
- g) Explain active learning. (2 marks)

[Total: 20 marks]

Question 13.2: Self-Training Rounds

The first model (trained on 300 grasps) finds 15,000 confident predictions. Round 2 adds 8,000 more. Validation accuracy on real labels is 0.84 before self-training, 0.87 after round 1 and 0.85 after round 2.

- a) Describe the steps of self-training. (3 marks)
- b) Calculate the size of the training set after round 1. (3 marks)
- c) Calculate the size of the training set after round 2. (3 marks)
- d) Interpret the validation results and decide when to stop. (3 marks)
- e) Explain confirmation bias in pseudo-labelling. (3 marks)
- f) Explain three ways to reduce confirmation bias. (3 marks)
- g) Explain what a confidence threshold of 0.95 means. (2 marks)

[Total: 20 marks]

Question 13.3: Other Semi-Supervised Methods

The team has force features and camera features for each grasp, and wants to try methods other than self-training.

- a) Explain label propagation. (4 marks)
- b) Explain how a similarity graph lets labels flow. (4 marks)
- c) Explain co-training using the force and camera views. (4 marks)
- d) Explain consistency regularisation. (4 marks)

- e) Explain why evaluation must use genuinely labelled data. (4 marks)

[Total: 20 marks]

Question 13.4: Pretext Tasks for IMU Data

The team has months of unlabelled IMU and vibration data and wants to pre-train a model before fine-tuning on a few fault labels.

- a) Define a pretext task. (2 marks)
- b) Explain next-value prediction. (3 marks)
- c) Explain masking. (4 marks)
- d) Explain contrastive learning. (3 marks)
- e) Explain cross-modal prediction and temporal-order tasks. (4 marks)
- f) Explain the two-stage recipe of pre-training and fine-tuning. (4 marks)

[Total: 20 marks]

Question 13.5: Twenty-Five Bearing Failures

The team has 25 labelled bearing failures and two years of unlabelled vibration logs, and must detect failing bearings.

- a) Explain the limitations of a purely supervised approach. (4 marks)
- b) Explain how semi-supervised learning could help. (3 marks)
- c) Explain how self-supervised learning could help. (3 marks)
- d) Explain how unsupervised anomaly detection could help. (4 marks)
- e) Explain why self-supervised pre-training needs a lot of unlabelled data. (2 marks)
- f) Recommend an approach and justify it. (4 marks)

[Total: 20 marks]

Week 14: Time Series and Forecasting

Question 14.1: Patterns in Pick-Task Demand

Hourly pick-task demand shows a slow rise over the year, a peak every morning shift, higher demand every Monday, busy periods during irregular sales campaigns, and minute-to-minute jitter.

- a) Explain the four components of a time series. (4 marks)
- b) Distinguish between additive and multiplicative decomposition. (4 marks)
- c) Identify each component in the demand data. (3 marks)
- d) Explain three methods of detecting seasonality. (3 marks)
- e) State where ACF peaks would appear for a weekly pattern in daily data. (2 marks)
- f) Define autocorrelation. (2 marks)
- g) Explain why time-series rows must not be shuffled. (2 marks)

[Total: 20 marks]

Question 14.2: Battery Capacity Fade

Monthly battery capacity is 120, 117, 115, 112 and 110 Ah. An ADF test on the raw series gives $p = 0.3$.

- a) Define stationarity. (3 marks)
- b) Explain the ADF test. (3 marks)
- c) Calculate the first differences. (3 marks)
- d) Calculate the average monthly decline. (3 marks)
- e) Explain seasonal differencing. (3 marks)
- f) Explain when a log transform is useful. (3 marks)
- g) Interpret the ADF p-value of 0.3. (2 marks)

[Total: 20 marks]

Question 14.3: Exponential Smoothing

Hourly motor temperatures are 60, 64 and 58 °C. Use $\alpha = 0.4$ and $s_1 = 60$.

- a) Calculate s_2 . (4 marks)
- b) Calculate s_3 and state the forecast for the next hour. (4 marks)
- c) Explain the effect of choosing a high or low α . (4 marks)
- d) Compare Holt and Holt-Winters smoothing. (4 marks)

- e) Choose a smoothing method for steady ambient temperature, slowly rising vibration, and hourly demand with a daily cycle. (4 marks)

[Total: 20 marks]

Question 14.4: ARIMA Forecasting

A vibration series follows the AR(1) model $x_t = 12 + 0.7x_{t-1}$. The last reading was 50.

- a) Explain p, d and q in ARIMA. (2 marks)
- b) Calculate the forecast for the next step. (3 marks)
- c) Calculate the forecast two steps ahead. (4 marks)
- d) Calculate the long-run mean. (3 marks)
- e) Explain how ACF, PACF and AIC help choose ARIMA orders. (4 marks)
- f) Explain the meaning of SARIMA(1,1,1)(1,1,1)₁₆₈ for hourly data. (4 marks)

[Total: 20 marks]

Question 14.5: When Will the Bearing Cross the Limit?

A robot's vibration_rms is 0.38 g and rising by about 0.005 g per day. The danger limit is 0.58 g.

- a) Calculate the number of days until the limit is reached. (4 marks)
- b) Explain how forecast uncertainty should affect the maintenance date. (3 marks)
- c) Explain walk-forward validation. (3 marks)
- d) Explain why forecasts must be compared with a naive baseline. (4 marks)
- e) Explain lag features. (2 marks)
- f) Propose a forecasting and maintenance plan for the fleet. (4 marks)

[Total: 20 marks]

Week 15: Deployment and MLOps — From Notebook to Robot Fleet

Question 15.1: Getting the Model Out of the Notebook

A fault model works well in a student's notebook. The maintenance team wants a daily risk list, the fleet system wants instant answers, and the robots need slip detection with no network.

- a) Explain batch deployment. (4 marks)
- b) Explain real-time API deployment. (4 marks)
- c) Explain streaming deployment. (3 marks)
- d) Explain edge deployment. (3 marks)
- e) Explain interactive app deployment. (2 marks)
- f) State what must be saved before deploying. (2 marks)
- g) Choose a deployment option for the daily risk list. (2 marks)

[Total: 20 marks]

Question 15.2: Apps and APIs

The team must build a demo for managers, a planners' dashboard and an API for the fleet system.

- a) Explain Streamlit and when to use it. (3 marks)
- b) Explain Gradio and when to use it. (3 marks)
- c) Explain Flask. (3 marks)
- d) Explain why FastAPI is popular for ML APIs. (3 marks)
- e) Explain how the API's "action" field turns a prediction into a prescription. (3 marks)
- f) Explain the role of Docker. (3 marks)
- g) Explain why models are converted to ONNX or TensorFlow Lite for robots. (2 marks)

[Total: 20 marks]

Question 15.3: The Decaying Model

A year after deployment, the fault model raises many false alarms. The company introduced heavier products, installed new motors that run hotter but fail less, and some thermistors now read 3 °C high.

- a) Explain data drift using this scenario. (4 marks)
- b) Explain concept drift using this scenario. (4 marks)

- c) Explain sensor drift using this scenario. (4 marks)
- d) Explain what should be monitored in production. (4 marks)
- e) Explain retraining with champion and challenger models. (4 marks)

[Total: 20 marks]

Question 15.4: CI/CD for the Fault Model

The team wants every code change tested automatically and new models released safely.

- a) Define MLOps. (2 marks)
- b) Explain the tests continuous integration should run. (3 marks)
- c) Explain continuous delivery and deployment. (4 marks)
- d) Explain continuous training. (3 marks)
- e) Explain how code, data and models are versioned. (4 marks)
- f) Explain shadow mode, canary release and rollback. (4 marks)

[Total: 20 marks]

Question 15.5: The End-to-End Pipeline

The team must design a complete MLOps pipeline for the fleet's predictive maintenance model.

- a) Describe the eight steps of the end-to-end pipeline. (4 marks)
- b) Explain how the pipeline links back to the four types of analysis. (3 marks)
- c) Explain how technician feedback becomes new labels. (3 marks)
- d) Explain the principles of responsible deployment for robot models. (4 marks)
- e) Explain why predictions should be explained to users. (2 marks)
- f) Design an MLOps pipeline for a learned charging model. (4 marks)

[Total: 20 marks]